PROJECT MADE SIMPLE

Preventing Cascading Failures in Urban Drainage Networks Using Graph Theory

A beginner-friendly guide to the problem, graph theory, simulation, strategies and findings

Project in one sentence

This project uses a simplified computer simulation to test whether choosing drainage pipes by both their failure risk and their importance to the wider network can protect water delivery better than choosing pipes by risk alone.

Use this guide to:

- Understand the project without reading all 38 pages at once.
- Explain the graph model and five maintenance strategies in plain English.
- Remember the main results and the limitations.

Based on the uploaded Graph Theory assignment report · October 2026

1. What problem is the project solving?

A drainage network collects rainwater and carries it through pipes and junctions to outlets. Each pipe can carry only a limited amount of water. If a pipe becomes blocked or overloaded, water may be redirected to other pipes. Those pipes may then become overloaded too. This chain reaction is called a **cascading failure**.

A simple example

1. Heavy rain More water enters the network.	**2. First pipe fails** A pipe is blocked or overloaded.	**3. Water reroutes** Other pipes receive extra flow.	**4. More failures** Some routes cannot cope.

Why does this matter?

• One failed pipe can affect more than its own location.

• Some pipes are more important because many routes to an outlet depend on them.

• Maintenance budgets are limited, so it is not possible to reinforce every pipe.

The main decision: If only a small number of pipes can be strengthened before a storm, which pipes should be selected?

The project does not build a real drainage system. It creates small synthetic networks in a computer and compares pipe-selection rules under controlled conditions.

2. How graph theory represents drainage

Graph theory helps represent a system as connected points and links. In this project, the graph is directed because water is assumed to move downhill.

Key terms

Graph term	Meaning in this project	Everyday picture
Node / vertex	A demand point, junction or outlet.	A place where pipes meet or water enters/leaves.
Edge	A pipe connecting two nodes.	A road connecting two places.
Direction	The permitted downhill flow direction.	A one-way street arrow.
Weight	A number attached to a pipe, used in routing or analysis.	A road's capacity or length.
Outlet	A destination where water can leave the network.	The exit at the end of a route.

What network did the report create?

• Five computer-generated networks were tested.

• Each network had 56 nodes and approximately 86–88 pipes.

• Each network had three outlets, with water directed downhill.

• The networks were designed to be plausible examples, not copies of a real city.

Remember: The graph is not just a picture. It lets the project calculate which pipes connect demand points to outlets and how a failure might affect other routes.

3. Research question and research gap

The report asks whether a maintenance strategy that combines two types of information can outperform a strategy that uses only individual pipe failure risk.

Two ideas being combined

• **Failure risk:** How likely is this particular pipe to fail?

• **Structural importance:** How important is this pipe for connecting demand areas to outlets?

Why is this worth testing?

Earlier work discussed in the report covers graph-based identification of critical pipes, cascading-failure models and risk-based maintenance. The report's stated gap is that, among the studies it reviewed, it did not find an equal-budget comparison of risk-only reinforcement against a combined risk-and-structure rule during cascades. The report is careful not to claim that no such study exists anywhere.

Research question

Can combining individual pipe risk with network structural importance reduce cascading damage more effectively than selecting pipes by risk alone?

What the project is trying to learn

• Does the combined strategy reduce the number of failed pipes?

• Does it help more water reach an outlet?

• Does performance change with rainfall intensity and the number of pipes reinforced?

• Which graph measure best identifies important pipes in this simulation?

4. The five strategies being compared

Every strategy is evaluated under the same conditions. When a budget allows reinforcement of K pipes, the strategies that select pipes all get the same K and the same assumed reinforcement benefit. This makes the comparison fairer.

ID	Strategy	How it chooses pipes	Why include it?
A	**Risk-based**	Chooses pipes with the highest individual failure risk.	Main baseline to beat.
B	**Graph-based**	Combines risk and directed edge betweenness.	Tests the project's main idea.
C	**Random**	Chooses pipes randomly.	Shows what random selection achieves.
D	**No intervention**	Reinforces no pipes.	Shows damage without prevention.
E	**Load-based**	Chooses pipes with the highest baseline load relative to capacity.	Tests whether load alone is enough.

What is directed edge betweenness?

In simple terms, it estimates how often a pipe lies on the shortest directed routes from demand points to outlets. A pipe used by many such routes may be structurally important. It is a graph-based indicator, not a direct measurement of real water pressure or flood risk.

The combined score gives weight to risk and structural importance. The selected setting was $\lambda = 0.75$, which places more weight on risk than on betweenness in the score used by this report.

5. How the simulation works

Think of the simulation as a repeatable storm experiment. The computer starts with a network, introduces possible blockages, routes water through the remaining pipes and checks whether any pipes become overloaded.

The process, step by step

1. Build the network

Create nodes, downhill pipes, demand points and outlets.

2. Set the storm

Choose a rainfall multiplier. Higher values mean more demand on the network.

3. Choose reinforcement

Each strategy selects pipes within its allowed budget.

4. Draw initial failures

Use the same random failure scenario for each strategy being compared.

5. Route the water

Water follows surviving downstream routes that can still reach an outlet.

6. Check overload

Any surviving pipe whose load exceeds its capacity fails in that round.

7. Repeat

Reroute water and check for new failures until no additional pipes fail.

8. Measure the outcome

Calculate failed pipes, unmet demand and other measures.

What does reinforcement mean in this model?

The report assumes that a reinforced pipe has 1.5 times its capacity and half its initial failure risk. These are illustrative modelling assumptions applied equally to all relevant strategies; they are not measured engineering values.

Important limitation: The water-routing rule is simplified and steady-state. It is not a hydraulically validated simulator and does not model all real-world processes such as storage, head loss, surcharge or storm timing.

6. How the comparison was kept fair

The report uses repeated trials because a single random failure pattern could make a strategy look unusually good or bad. Repeating the experiment gives a more stable comparison.

Conditions tested

- **Networks:** 5 synthetic networks.
- **Rainfall multipliers:** 0.7, 1.0, 1.3 and 1.6.
- **Reinforcement budgets:** 5%, 10% and 20% of pipes.
- **Test trials:** 100 per network for each rainfall-budget condition (500 paired trials per condition).
- **Fair comparison:** strategies use the same random failure scenarios.
- **Separate tuning and testing:** one set of trials selects λ; a held-out test set is used for headline results.

The key outcome measures

Measure	Plain meaning	Better
Failure rate	Share of pipes that have failed at the end.	Lower
Unmet demand	Share of demand at nodes with no surviving path to an outlet.	Lower
Retained capacity	Original capacity belonging to surviving pipes, relative to total.	Higher
Secondary failures	Failures that happen after the initial blockages.	Lower
Largest affected region	Size of the largest group of cut-off nodes.	Lower

The report also uses paired statistical tests and confidence intervals to judge whether observed differences are consistent enough to count as strong evidence, rather than relying only on average values.

7. What did the project find?

The result is mixed, and that is an important part of the conclusion. Strategy B did not reliably reduce the proportion of failed pipes compared with Strategy A. However, it generally reduced unmet demand in the simulated networks.

Main results at a glance

Question	Finding from the report
Did graph-based Strategy B significantly reduce failure rate versus risk-only A?	No. None of the 12 rainfall-budget conditions showed a significant reduction after multiple-testing correction. Changes ranged from –0.5% to +2.9%.
Did it reduce unmet demand?	Yes in 11 of 12 conditions. Reductions ranged from 2.1% to 18.6%.
Where was the largest unmet-demand gain?	At the 10% reinforcement budget and rainfall multiplier 0.7: 18.6% relative reduction.
What happened as rainfall rose?	Damage increased for every strategy, and Strategy B's advantage generally became smaller.
What did the graph measure show?	Betweenness was moderately related to single-pipe impact in the model (Spearman correlation $\rho = 0.42$). Risk and load were nearly uncorrelated with that impact measure.

One useful example

At rainfall multiplier 1.0 and a 10% budget, mean unmet demand was about 0.169 for risk-based Strategy A and 0.146 for graph-based Strategy B. That is a decrease of about 0.023, or 2.3 percentage points of demand, equal to a 13.5% relative reduction in this simulation.

Do not simplify this to 'graph theory always wins'. The strongest supported conclusion is that it helped protect delivery in this model, but did not reliably reduce the number of failed pipes.

8. What should we be careful about?

The results are useful for comparing strategies inside the experiment, but they are not proof that the same improvements would happen in a real city.

Main limitations

• The networks are synthetic, small and not validated representations of real drainage systems.

• The routing algorithm is a simplified approximation, not a hydraulic simulation.

• Parameter values such as failure risk, capacity margins and reinforcement effects are illustrative assumptions.

• Only pipes can fail; the model excludes junction and pump failures, storage and other real-world processes.

• Unmet demand depends on whether a surviving path to an outlet exists, so this measure partly reflects the model's own graph structure.

What would improve the project next?

• Use real drainage network data.

• Compare simulated rankings and cascades against a hydraulic simulator such as SWMM.

• Use rainfall-runoff inputs and storms that change over time.

• Include other failure types and features such as storage and surcharge.

• Test more direct optimisation methods instead of relying only on a ranking score.

The honest interpretation

This is a controlled graph-theory and simulation study. It demonstrates a way to compare maintenance rules, highlights a potential benefit for water delivery, and identifies what must be validated before any real-world use.

9. The whole project in one page

A short explanation you can use in a presentation

“My project studies cascading failures in urban drainage networks. A pipe failure can redirect water to other pipes, which may overload and fail too. I represent a drainage network as a directed graph: junctions and outlets are nodes, and pipes are directed edges. I create five synthetic networks and compare five strategies for choosing pipes to reinforce. The main strategy combines each pipe's failure risk with directed edge betweenness, a measure of how important the pipe is to routes from demand points to outlets. I test the strategies under different rainfall levels and reinforcement budgets, using the same random failure scenarios to make comparisons fair. The graph-based strategy did not significantly reduce the failed-pipe rate compared with risk-only maintenance, but it reduced unmet demand in 11 out of 12 tested conditions. Since the networks and routing model are simplified, the results need validation with real drainage data and a hydraulic simulator before they can support real-world flood-prevention decisions.”

Quick glossary

Term	Remember it as
Cascade	One failure triggers additional failures.
Directed graph	Connected points with arrows showing allowed direction.
Edge betweenness	How often a pipe appears on selected routes through the network.
Risk-based	Prioritise pipes most likely to fail.
Unmet demand	Demand that cannot reach an outlet in the model.
Reinforcement budget	The limited number of pipes that can be strengthened.
Synthetic network	A computer-generated example, not a real city's network.

Three things to remember: (1) The problem is a chain reaction of pipe failures. (2) Strategy B combines risk and network position. (3) The main benefit was lower unmet demand in the simulation—not a proven reduction in real-world flooding.

Source: Simplified from the uploaded assignment report, “Preventing Cascading Failures in Urban Drainage Networks Using Graph Theory” (October 2026).